Gelivi Sai Chaitanya

Norman, OK | chaitanyagelivi03@gmail.com | (405) 413-2430 | Github | Portfolio

Education

University of Oklahoma Norman, OK
Master of Science in Computer Science GPA: 3.91/4.00 *Aug 2024 – May 2026*
Relevant Coursework: Advanced Algorithms, Database Systems, Machine Learning, Computer Networks, Software Engineering, Distributed Systems, Computer Vision

Vardhaman College of Engineering Hyderabad, India
Bachelor of Technology in Computer Science (AI & ML) GPA: 9.66/10.00 *Dec 2020 – May 2024*

Professional Experience

MassMutual Remote
Machine Learning Intern *Aug 2022 – Jan 2023*

- Built and deployed ARIMA/SARIMA and LSTM time-series models on 3+ years of transactional data (100k+ records across timestamps, product categories, and geography), improving monthly ticket volume forecast accuracy by 20%.
- Engineered ETL pipelines for multi-dimensional data ingestion, cleaning, and feature extraction, cutting preprocessing time by 30% and enabling faster model iteration cycles.
- Delivered interactive Matplotlib dashboards and executive reports translating model outputs into actionable operational decisions for resource planning teams.

Technical Skills

Languages: Python, Java, C/C++, JavaScript, SQL, Dart, HTML5, CSS3
Frameworks & Libraries: React, Node.js, Express.js, Flutter, Ethers.js, TensorFlow, Keras, Scikit-learn, Pandas, NumPy
Databases: PostgreSQL, MySQL, MongoDB, Oracle, Cloud Firestore
Tools & Platforms: Git, Docker, Linux, Hardhat, Firebase, Postman, PyInstaller, Jupyter
Concepts: Data Structures & Algorithms, OOP, REST APIs, Smart Contracts, Cryptography, CI/CD

Projects

Postgres Native Vector Retrieval | *C, Python, PostgreSQL, pgvector, HNSW, PL/pgSQL, Docker*

- Authored a native PostgreSQL extension in C implementing HNSW-based approximate nearest neighbor search directly inside the database engine, eliminating the need for an external vector store such as Pinecone or Weaviate and reducing infrastructure complexity by removing a network hop from every query.
- Achieved sub-10ms semantic query latency on 100k+ embedding datasets with 768-dimensional vectors, delivering a 5–10x speedup over a baseline brute-force kNN scan while maintaining recall@10 above 0.95.
- Benchmarked retrieval performance across multiple HNSW index configurations (graph connectivity and search depth parameters) and containerized the full stack with Docker Compose, enabling one-command reproducibility and cutting evaluation setup time from hours to minutes.

TrackR — Full-Stack Job Application Tracker | *React, Node.js, MongoDB, JWT, Google OAuth 2.0* GitHub | Live

- Built and deployed a full-stack MERN application tracking 10+ fields per job application with real-time search, filtering, CSV export, and dark mode; shipped to production on Vercel and Render with MongoDB Atlas as the database, serving a live public URL.
- Implemented hybrid authentication supporting both email/password (bcrypt-hashed credentials with JWT-based sessions) and Google OAuth 2.0, with automatic account linking via shared email matching so users can seamlessly switch between sign-in methods without losing their data.
- Designed the analytics dashboard using MongoDB aggregation pipelines ($group, $match, $cond) to compute response rates and status breakdowns server-side rather than pulling documents to the client, reducing payload size and keeping API responses consistent as the dataset scales.

Credit Card Fraud Detection | *Python, Scikit-learn, Hidden Naive Bayes, Bayesian Belief Networks*

- Implemented Hidden Naive Bayes and Bayesian Belief Network classifiers on a highly imbalanced real-world financial transaction dataset (fraud cases under 1% of total volume), applying stratified sampling and decision threshold tuning to maximize F1-score on the minority fraud class.
- Evaluated both models against a standard Naive Bayes baseline using precision, recall, F1-score, and ROC-AUC; HNB delivered higher precision on fraudulent transactions, directly reducing false negatives — the most costly error class in financial fraud detection.
- **Research published at IEEE I2CT 2024** (peer-reviewed, 200+ attendees); demonstrated practical applicability of probabilistic graphical models for anomaly detection in production-scale financial systems. DOI: 10.1109/I2CT61223.2024.10544328

Leadership & Honors

- **Vice President**, Competitive Coding Club, Vardhaman College of Engineering (2022–2024) — grew participation by 40%; recipient, Best Club Activities Award, IEEE Envision 2024.
- **Cultural Committee Member**, Indian Student Association (ISA), University of Oklahoma (2024–Present).